

CloudShift - Software Management | Test Plan and Evidence

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1. Test Strategy & Scope

The testing strategy validates that all 9 applications migrate cleanly to the cloud with zero data corruption, full blackout compliance, and instant rollback execution if cutover failures occur. It enforces rigorous validation across functional requirements, boundary conditions, decision matrices, defect logs, and project quality metrics derived from empirical case-study evidence.

2. System Test Cases for Functional Requirements (FR-01 to FR-12)

Table 2A: Functional Requirement Test Cases (FR-01 to FR-06)

Test Case ID	Requirement	Test Objective	Test Input / Action	Expected Result	Pass/Fail Criteria
TC-FR-01	FR-01 (CDC Sync)	Validate real-time CDC replication lag between DBs	Insert 5,000 DB records into on-prem DB	Target cloud DB mirrors all committed records and reaches zero replication lag before cutover	PASS if all committed transactions are present and cutover lag is 0
TC-FR-02	FR-02 (Schema Trans)	Verify automated schema mapping accuracy	Run schema transformation tool	100% table schemas, constraints, and indexes migrated	PASS if 0 schema errors
TC-FR-03	FR-03 (DNS Reroute)	Test DNS cutover and load balancer swap	Execute DNS update script	Traffic routes to the cloud target after the approved DNS/load-balancer change	PASS if approved cloud target receives production traffic
TC-FR-04	FR-04 (Blackout Lock)	Test deployment pipeline locking during blackout	Trigger migration script on Working Day 19 of a 22-working-day month	Execution blocked with error message: "Blackout Active"	PASS if execution blocked
TC-FR-05	FR-05 (Session Migr)	Validate active user session migration to Redis	User logged in on-prem, trigger session sync	User remains authenticated post-cutover without login	PASS if session active
TC-FR-06	FR-06 (IaC Provision)	Test Terraform script provisioning	Run terraform apply	VPC, Subnets, Security Groups, and K8s cluster created	PASS if 0 IaC errors

Table 2B: Functional Requirement Test Cases (FR-07 to FR-12)

Test Case ID	Requirement	Test Objective	Test Input / Action	Expected Result	Pass/Fail Criteria
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TC-FR-07	FR-07 (Config Mgmt)	Verify dynamic secret injection	Launch app container with parameter store	DB credentials loaded securely from parameter store	PASS if app connects
TC-FR-08	FR-08 (IAM & RBAC)	Validate SSO RBAC authorization	Log in as Non-Admin & Admin	Non-Admin denied migration triggers; Admin allowed	PASS if RBAC enforced
TC-FR-09	FR-09 (Synthetic Mon)	Test automated 30s health check bot	Inject simulated 500 error in staging	Synthetic monitor detects failure within ≤ 30 s	PASS if alert fired
TC-FR-10	FR-10 (Auto Rollback)	Verify automated DNS rollback execution	Fail health check post-cutover	DNS automatically reverted to on-prem within ≤ 15 min	PASS if RTO ≤ 15 min
TC-FR-11	FR-11 (Audit Log)	Validate append-only audit trail	Perform cutover action	Audit log records event, timestamp, user ID, SHA256 hash	PASS if log immutable
TC-FR-12	FR-12 (Dashboard)	Verify management UI dashboard metrics	Open dashboard during CDC sync	Displays real-time replication lag, blackout timer, status	PASS if metrics match

3. Boundary Value Analysis (BVA) for Month-End Blackout Window

3.1 Field Specification

- **Input Variable:** Working Day of the Month ($W \in [1, 22]$)
- **Blackout Window Policy:** Last 5 working days of the month ($W \in [18, 22]$) are BLACKOUT DAYS (cutover disabled). Days $W \in [1, 17]$ are PERMITTED DAYS (cutover enabled).

3.2 BVA Test Cases Table

BVA Case ID	Working Day Value (W)	Boundary Classification	Expected System Behavior	Test Result
BVA-01	$W = 0$	Below Min (Invalid)	Input Error / Rejected	PASS
BVA-02	$W = 1$	Min Valid Working Day	Cutover ALLOWED (Normal Operation)	PASS
BVA-03	$W = 2$	Min + 1	Cutover ALLOWED (Normal Operation)	PASS
BVA-04	$W = 16$	Nominal Non-Blackout Day	Cutover ALLOWED (Normal Operation)	PASS
BVA-05	$W = 17$	Boundary: Last Permitted Day (18 - 1)	Cutover ALLOWED (Normal Operation)	PASS
BVA-06	$W = 18$	Boundary: First Blackout Day (Min Blackout)	Cutover BLOCKED (Blackout Lock Active)	PASS
BVA-07	$W = 19$	Inside Blackout Window	Cutover BLOCKED (Blackout Lock Active)	PASS
BVA-08	$W = 21$	Inside Blackout Window	Cutover BLOCKED (Blackout Lock Active)	PASS

BVA-09	W = 22	Boundary: Last Blackout Day (Max Blackout)	Cutover BLOCKED (Blackout Lock Active)	PASS
BVA-10	W = 23	Above Max Working Days (Invalid)	Input Error / Rejected	PASS

4. Equivalence Partitioning (EP) for Migration Window

Partition ID	Working Day Range	Partition Class	System Expected Behavior
EP-01	$W < 1$	Invalid Low	System Rejects Request (Invalid Day Error)
EP-02	$1 \leq W \leq 17$	Valid Permitted Window	System Permits Migration & Cutover
EP-03	$18 \leq W \leq 22$	Valid Blackout Window	System Locks Cutover Pipeline (Blackout Enforcement)
EP-04	$W > 22$	Invalid High	System Rejects Request (Invalid Day Error)

5. Decision Table for Cut-over Go/No-Go Rules

5.1 Condition Definitions

- **C1 (Checksum Match):** Pre-cutover DB SHA-256 checksum parity = 100%.
- **C2 (Blackout Active):** Current calendar date falls within last 5 working days of month ($W \in [18, 22]$).
- **C3 (Health Check Pass):** Synthetic test returns HTTP 200 & API P95 latency ≤ 200 ms.
- **C4 (Rollback Verified):** Automated rollback target and DB backup verified.

5.2 Decision Matrix

Rule ID	C1 Checksum 100%	C2 Blackout Active	C3 Health Check Pass	C4 Rollback Verified	Decision Outcome / Action
R-01	True	False	True	True	GO: Execute Traffic Cutover to Cloud
R-02	True	True	True	True	NO-GO: Defer Cutover (Blackout Window Active)
R-03	False	False	True	True	NO-GO: Abort Cutover (Data Checksum Mismatch)
R-04	True	False	False	True	NO-GO: Trigger Automated Rollback to On-Prem
R-05	True	False	True	False	NO-GO: Abort Cutover (Rollback Plan Unverified)
R-06	False	True	False	False	NO-GO: Abort Cutover (Multiple Safety Failures)

6. Comprehensive Defect Log (Pre-Release & Post-Release Evidence)

6.1 Pre-Release Dry-Run & Staging Defect Evidence (E_pre = 18 defects)

Table 6A: Pre-Release Defects (DEF-01 to DEF-09)

Defect ID	Associated App	Summary / Description	Severity	Discovery Phase	Status	Root Cause & Resolution
DEF-01	App1	CDC replication buffer overflow during load test	Major	Staging Sync	Closed	Buffer size increased from 128MB to 512MB
DEF-02	App2	DNS TTL propagation delayed beyond 5 minutes	Medium	Wave 1 Cutover	Closed	TTL lowered from 86400s to 60s in Route53
DEF-03	App7	Legacy hardcoded DB connection string crashed app	Critical	Wave 3 PERT	Closed	Replaced with environment parameter injection
DEF-04	App4	Blackout Guard allowed trigger on 18th working day	Critical	Integration Test	Closed	Corrected boundary operator from > to >=
DEF-05	App3	Redis session token missing expiration header	Low	Wave 1 Staging	Closed	Added default 24h TTL policy in Redis
DEF-06	App8	Deprecated PHP 5.6 module incompatible with cloud OS	High	Reverse Eng	Closed	Containerized legacy PHP runtime in Docker
DEF-07	App5	P95 latency reached 340ms under 1,000 req/s	Medium	Load Testing	Closed	Added read-replica DB & index tuning
DEF-08	App9	Missing table primary key caused CDC checksum failure	High	Data Validation	Closed	Created synthetic unique index on target schema
DEF-09	App6	Automated rollback script timed out at 18 minutes	Critical	DR Simulation	Closed	Parallelized container shutdown (RTO now 8m)

Table 6B: Pre-Release Defects (DEF-10 to DEF-18)

Defect ID	Associated App	Summary / Description	Severity	Discovery Phase	Status	Root Cause & Resolution
DEF-10	App2	Audit logger dropped logs during network partition	Low	Security Audit	Closed	Implemented local disk queue for log spooling
DEF-11	App1	SSL/TLS certificate handshake mismatch on ALB	Medium	Security Testing	Closed	Updated IAM certificate ARN binding
DEF-12	App3	Heap memory leak during 24-hour continuous CDC sync	Major	Endurance Test	Closed	Fixed unclosed DB connection pool handle

DEF-13	App4	Dynamic secrets parameter fetch timeout on cold start	High	Integration Test	Closed	Added SDK retry policy with exponential backoff
DEF-14	App5	Concurrent session token collision under traffic spikes	Medium	Performance Test	Closed	Updated session token generator to UUIDv4
DEF-15	App6	Synthetic health check bot false positive alert	Low	Monitoring Test	Closed	Adjusted synthetic probe retry threshold to 3
DEF-16	App7	Legacy hardcoded file path missing on Linux instance	Critical	Staging Execution	Closed	Refactored path to S3 object bucket adapter
DEF-17	App8	Missing DB index on invoice timestamp query	Medium	DB Audit	Closed	Created B-tree index on invoice_created_at
DEF-18	App9	System daemon failed to auto-restart on reboot	High	System Test	Closed	Configured systemd unit file with auto-restart

6.2 Post-Migration Warranty Defects (E_post = 2 defects)

Table 6C: Post-Migration Defects

Defect ID	Associated App	Summary / Description	Severity	Discovery Phase	Status	Root Cause & Resolution
POST-01	App7	Minor UI alignment artifact on legacy report page	Low	Post-Migration Warranty	Closed	CSS rule updated in hotfix patch
POST-02	App2	Intermittent 503 on legacy batch export route	Minor	Post-Migration Warranty	Closed	Increased container worker thread limit

7. Quality Metrics Computation

7.1 Input Metrics Log Data

- **Pre-Release Dry-Run Defects (E_pre):** 18 defects
- **Post-Migration Warranty Defects (E_post):** 2 minor defects
- **Total Applications (N_apps):** 9 applications (App1 = 8d, App2 = 10d, App3 = 6d, App4 = 12d, App5 = 9d, App6 = 15d, App7 = 20d, App8 = 18d, App9 = 22d)
- **Total Project Effort (E_total):** 126 person-days

7.2 Defect Removal Efficiency (DRE) Calculation

Formula: $DRE = (E_{pre} / (E_{pre} + E_{post})) \times 100\%$

Calculation: $DRE = (18 / (18 + 2)) \times 100\% = (18 / 20) \times 100\% = 90.0\%$

7.3 Defect Density Calculations

1. Defect Density per Application:

Formula: $\text{Defect Density}_{\text{app}} = (E_{\text{pre}} + E_{\text{post}}) / N_{\text{apps}}$

Calculation: 20 defects / 9 applications = **2.22 defects/application**

2. Defect Density per Person-Day of Migration Effort:

Formula: $\text{Defect Density}_{\text{effort}} = (E_{\text{pre}} + E_{\text{post}}) / E_{\text{total}}$

Calculation: 20 defects / 126 person-days = **0.159 defects/person-day**

8. Evidence Interpretation

The defect log contains 18 pre-release defects and 2 post-migration warranty defects. The resulting DRE (90.0%) and defect-density figures (2.22 defects/app, 0.159 defects/person-day) are calculated directly from the supplied empirical test-log evidence. They should be presented as case-study evidence, not as independently audited production metrics.

The BVA/EP model deliberately treats working days 18–22 as the blackout partition because the case defines the last 5 working days of a 22-working-day month as unavailable for migration/cutover.

9. Conclusion

The CloudShift migration test suite successfully demonstrates comprehensive QA coverage across all 9 target applications. Functional requirements FR-01 through FR-12 have been rigorously validated. Edge cases and month-end blackout policies are properly enforced via Boundary Value Analysis and Equivalence Partitioning models, while Go/No-Go decision rules safeguard data integrity and system availability. With a Defect Removal Efficiency of 90.0%, the migration framework meets enterprise software engineering and academic compliance standards.